Tyler Jay Yang

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Technical Skills

Languages: Python, C, C++, Golang, OCaml, SML, Java, SQL, MATLAB, Julia, R
Other: Git, Shell Scripting, Vim, AWS (EC2, DynamoDB, S3), Docker, MPI, SLURM

Experience

Software Engineer May 2026 – Aug 2026
Aven Financial

Technical Advisor | Scale AI Aug 2025 – Dec 2025
Agentic AI workflow evaluation, AI safety and alignment Python | C++ | Docker

Research Assistant | CMU Mechanical AI Laboratory Jun 2025 – Jul 2025
Multimodal AI for reaction extraction from chemical literature Python | PyTorch | HuggingFace | Ray

Projects

Mangrove: Optimal Subgraph Execution Schedules Apr 2026 – May 2026
Size-Aware Tensor Scheduler Python | C++

- Engineered a standalone C++ hardware simulator and decoupled DAG verifier to strictly enforce memory feasibility and structurally validate schedules against an analytical roofline model.
- **Reduced execution latency by up to 47%** on complex benchmarks using an exact dynamic programming solver for subgraph routing, scaling to larger graphs via A* beam search, simulated annealing, and greedy chain-fusion heuristics.

Distributed Training Infrastructure Jan 2026 – Apr 2026
Machine Learning Systems Python | NumPy | CUDA | TIRx | MPI

- Engineered GEMM kernels for NVIDIA Blackwell (SM100) GPUs using TIRX DSL, scaling throughput to **1,322 TFLOP/s.**
- Maximized hardware utilization via TMA asynchronous memory loads, warp specialization, 4-stage software pipelining, and 2-CTA cluster multi-casting.
- Built distributed training communication protocols (All-Reduce, All-Gather, and Reduce-Scatter) from scratch via MPI.

Optimizing C-to-x86 Compiler Jan 2026 – May 2026
Compiler Design OCaml | C | LLVM | x86 ASM

- Developed a multi-pass compiler translating a complex subset of C into optimized x86-64 assembly and LLVM IR, enabling modular extensibility and multi-target compilation.
- **Achieved up to a 2x speedup over GCC -O1** across 19 core benchmarks through aggressive optimization passes—including Sparse Conditional Constant Propagation (SCCP), Partial Redundancy Elimination (PRE)

AWAP 2026 Matchmaking Server Dec 2025 – Jan 2026
Backend Engineering for ACM at CMU, Algorithms with a Purpose (AWAP) Hackathon Golang | Docker | S3 | EC2 | DynamoDB

- Coordinated a team of 5 developers to redesign the backend for AWAP 2026
- **Reduced memory usage by 30%** over AWAP 2025 backend using semaphore-throttled worker pool

Multimodal Framework for Federated Learning (MuFFLe) Jan 2026
Team Lead + Most Collaborative Award, CMU x NVIDIA Federated Learning Hackathon Python | S3 | NVFlare

- Prototyped extensible federated learning infrastructure for data harmonization across different biobanks
- Designed secure methods for aggregation of sensitive, multimodal data for deep neural network training

MolSnap Oct 2025
Most Technical Project Award, St. Jude Knowledge in Data Science BioHackathon Python | PyTorch

- Architected fine-tuning pipeline for converting molecular structures to SMILES strings
- **Increased accuracy by 16%** for complex macrocycle structures over SoTA baseline

Education

Carnegie Mellon University, School of Computer Science Pittsburgh, PA
B.S. in Computer Science, Computer Systems Concentration, GPA: 3.90 May 2028

Teaching Assistant, 15-210 Parallel Data Structures and Algorithms Spring 2026
Teaching Assistant, 21-259 Multivariable Calculus Fall 2025
Board Member, CMU Computational Biology Society Fall 2025